



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/24

Paper 2 Further Pure Mathematics 2

May/June 2026

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

- 1 (a) Find the set of values of k for which the system of equations

$$3x + ky + 7z = 1,$$

$$8x + y + 2z = 2,$$

$$8x + 2y + 4z = 3,$$

has a unique solution and interpret this situation geometrically.

[4]

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- (b) Given that $k = \frac{7}{2}$, show that the system of equations in part 1(a) is inconsistent and interpret this situation geometrically.

[2]

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2 Let $\mathbf{A} = \begin{pmatrix} a & -1 \\ 5 & 1 \end{pmatrix}$, where a is a constant. It is given that 2 is an eigenvalue of $\mathbf{A}$.

(a) Find a and the other eigenvalue of $\mathbf{A}$.

[4]

[illegible]



(b) Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A}^5 = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$.

[5]

[illegible]



3 Find the particular solution of the differential equation

$$\frac{dy}{dx} + y = e^{\frac{1}{2}x},$$

given that $y = 3$ when $x = 0$. Give your answer in the form $y = f(x)$.

[7]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



4 The curve C has parametric equations

$$x = 2 \cosh 2t + 3t, \quad y = \frac{3}{2} \cosh 2t - 4t, \quad \text{for } 0 \leq t \leq 1.$$

(a) Find, in terms of e , the length of C .

[5]

[illegible]



(b) Show that

$$\frac{d^2y}{dx^2} = \frac{a \cosh 2t}{(b \sinh 2t + c)^n},$$

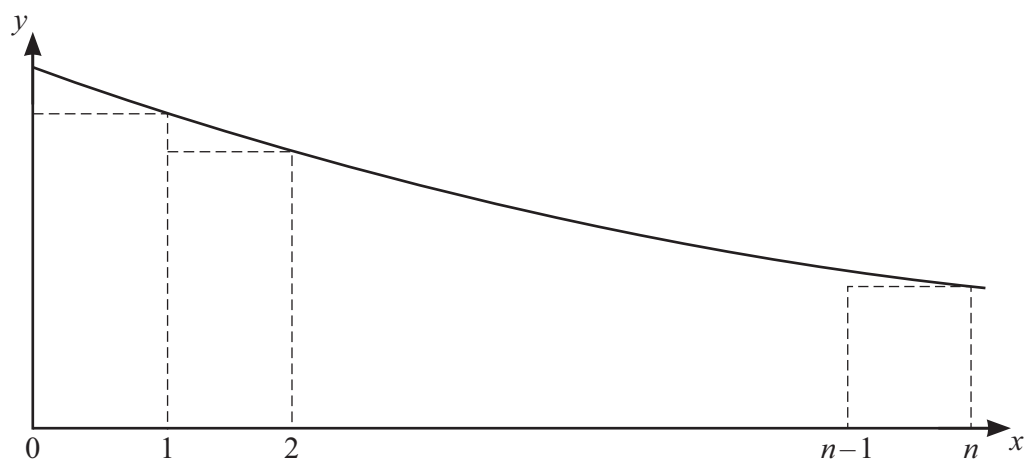
where a, b, c and n are integers to be determined.

[5]





5



The diagram shows the curve $y = \frac{1}{\sqrt{4x^2 + 4x + 3}}$ for $x \geq 0$, together with a set of n rectangles of unit width. By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{\sqrt{4r^2+4r+3}} < \frac{1}{2} \ln \left(\frac{1+2n+\sqrt{4n^2+4n+3}}{1+\sqrt{3}} \right). \quad [9]$$

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9

[illegible]



- 6 Let $I_n = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} e^x \cos^n x \, dx$, where n is a non-negative integer.

(a) Show that, for $n \geq 2$,

$$I_n = n(n-1) \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} e^x \cos^{n-2} x \sin^2 x \, dx - nI_n,$$

and deduce the reduction formula

$$(n^2 + 1)I_n = n(n - 1)I_{n-2}. \quad [6]$$

[illegible]



(b) Find the exact value of I_4 .

[4]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.



$$\sec 7\theta = \frac{\sec^7 \theta}{64 - 112 \sec^2 \theta + 56 \sec^4 \theta - 7 \sec^6 \theta}. \quad [6]$$

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(b) Hence obtain the roots of the equation

$$x^7 - 14x^6 + 112x^4 - 224x^2 + 128 = 0$$

in the form $\sec q\pi$, where q is rational.

[5]





8 It is given that $v = xy^3$ and

$$xy^2 \frac{d^2y}{dx^2} + 2xy \left(\frac{dy}{dx} \right)^2 + (3x+2)y^2 \frac{dy}{dx} + (x+1)y^3 = x+5.$$

(a) Show that

$$\frac{1}{3} \frac{d^2v}{dx^2} + \frac{dv}{dx} + v = x+5. \quad [4]$$

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(b) Find the general solution for y in terms of x . [7]

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(c) Deduce that, for large positive values of x and for any initial conditions,

$$y \approx \sqrt[3]{a + \frac{b}{x}},$$

where the constants a and b are to be determined. [2]

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Additional page

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